



**Northeastern University**  
College of Engineering

# Renewable Energy Investment advisor for campuses

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Project Proposal

EMGT 5220 Engineering Project Management (Section – 4)  
Spring 2026

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## TABLE OF CONTENTS

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|                                               |    |
|-----------------------------------------------|----|
| LETTER OF TRANSMITTAL .....                   | i  |
| EXECUTIVE SUMMARY .....                       | ii |
| 1.0 INTRODUCTION.....                         | 1  |
| 1.1 Problem.....                              | 1  |
| 1.2 Solution .....                            | 1  |
| 2.0 PURPOSE AND OBJECTIVES.....               | 2  |
| 2.1 Purpose .....                             | 2  |
| 2.2 Objective.....                            | 2  |
| 3.0 TECHNICAL OVERVIEW .....                  | 3  |
| 4.0 IMPLEMENTATION PLAN .....                 | 4  |
| 4.1 WBS.....                                  | 4  |
| 4.2 Schedule.....                             | 4  |
| 4.3 Responsibility Chart .....                | 4  |
| 4.4 Resource Allocation .....                 | 4  |
| 4.5 Stakeholders .....                        | 4  |
| 5.0 EXECUTION PLAN .....                      | 5  |
| 5.1 Project Monitoring.....                   | 5  |
| 5.2 Project Control.....                      | 5  |
| 5.3 Project Audit .....                       | 5  |
| 5.4 Project Closure.....                      | 5  |
| 6.0 RISK ASSESSMENT MANAGEMENT PLAN .....     | 6  |
| 6.1 Identification and Analysis of Risks..... | 6  |
| 6.2 Risk Management Plan .....                | 6  |
| 6.3 Continuous Risk Assessment .....          | 6  |
| 7.0 FINANCIAL PLAN WITH BUDGET .....          | 7  |
| 7.1 High-Level Details .....                  | 7  |
| 7.2 Budget Justification .....                | 7  |
| 8.0 TEAM CREDENTIALS.....                     | 8  |
| APPENDICES.....                               | 9  |
| APPENDIX A: Work Breakdown Structure .....    | 9  |
| APPENDIX B: Project Schedule.....             | 9  |
| APPENDIX C: RACI Matrix.....                  | 9  |
| APPENDIX D: Resource Allocation .....         | 9  |

## LETTER OF TRANSMITTAL

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17th April 2026  
Prof. Himlona Palikhe  
259 Snell Engineering  
360 Huntington Avenue  
Boston, MA 02115

Dear Prof. Himlona Palikhe,

Please find enclosed the project proposal for our Renewable Energy Investment Advisor (REIA), developed to address the absence of a structured, data-driven framework for evaluating renewable energy investment options at Northeastern University. As the university pursues its sustainability commitments, capital allocation decisions for energy infrastructure are too often made without a consistent, integrated basis for comparison. This framework is designed to change that.

The REIA offers a modular, Excel-based decision-support environment that evaluates solar photovoltaic, wind, geothermal, and battery storage investments across four analytical dimensions: campus energy demand forecasting, renewable energy production estimation, cost-benefit analysis, and environmental impact assessment. Rather than prescribing a specific technology configuration, the framework enables transparent, scenario-based comparison of investment alternatives before capital resources are committed — improving decision quality, reducing uncertainty, and supporting alignment across institutional stakeholders.

This proposal outlines the technical architecture of the framework, a phased implementation plan spanning 22 weeks from May through October 2026, a full Work Breakdown Structure and project schedule, a RACI responsibility matrix, a comprehensive resource and budget plan, and a risk management strategy.

Our team brings complementary expertise across engineering management, data systems, financial analysis, and sustainability assessment, and we are confident that the REIA delivers a meaningful and replicable contribution to Northeastern's energy planning capabilities.

Thank you for your time and consideration. We welcome any questions or feedback you may have.

Sincerely,

Lokesh, Ayush  
Nadukuru, Jyoti Swarup Rao  
Senthilvel, Sujeeth  
Sudhandrakumar, Santhosh Kumar

## EXECUTIVE SUMMARY

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The Renewable Energy Investment Advisor project proposes the development of a structured, data-driven decision-support framework to optimize energy investment decisions for Northeastern University. As institutions pursue aggressive sustainability and carbon reduction goals, they are increasingly required to evaluate multiple renewable energy alternatives under financial, operational, and environmental constraints. Currently, these decisions are often made using fragmented analyses, leading to uncertainty, suboptimal capital allocation, and increased financial risk. This project addresses this gap by developing an Integrated Renewable Energy Investment Optimization Model that enables portfolio-level evaluation of energy investment options. The framework consolidates key decision criteria—including cost, feasibility, environmental impact, and risk—into a unified analytical model, allowing stakeholders to compare multiple investment scenarios before committing capital. The model evaluates combinations of renewable technologies such as solar photovoltaic, wind, geothermal systems, and battery storage, and generates decision-ready outputs including cost-benefit comparisons, emissions reduction estimates, and scenario-based insights.

The core of the solution is an Excel-based analytical framework supported by structured modeling and scenario analysis techniques. The system integrates energy demand forecasting, renewable energy generation estimation, cost-benefit analysis, and environmental impact assessment into a single pipeline. Scenario analysis and sensitivity testing allow decision-makers to evaluate trade-offs and understand how changes in assumptions affect financial and sustainability outcomes. The final deliverables include an optimization-ready analytical model, scenario comparison outputs, and a structured decision report to support investment planning. The project follows a structured implementation plan spanning approximately 26 weeks, from May 2026 to November 2026, covering initiation, planning, execution, integration, validation, and closure phases. The execution phase represents the most resource-intensive portion of the project, focusing on model development, integration, and scenario analysis.

The total project effort is estimated at approximately 2,451 hours, with a total labor cost of \$146,350, reflecting the analytical and knowledge-intensive nature of the framework. Labor accounts for over 93% of the total budget, with minimal material costs, as the project primarily involves data processing, modeling, and decision-support development. A contingency reserve is included to manage uncertainties related to data availability and model refinement. The project team consists of roles spanning project management, data and systems analysis, financial modeling, and sustainability assessment, supported by institutional stakeholders, technical experts, and external contributors. This structure ensures that the framework is both technically robust and aligned with real-world decision-making requirements. From a financial perspective, the framework delivers value by improving investment efficiency and reducing the risk of suboptimal capital allocation. Even small improvements in decision quality can lead to significant long-term cost savings.

In conclusion, the Renewable Energy Investment Advisor provides a structured and data-driven approach to optimize energy investments, enabling cost-effective, sustainable, and strategically aligned decision-making.

## **1.0 INTRODUCTION**

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### **1.1 Problem**

The problem addressed by this project is the absence of a structured and integrated approach for evaluating and prioritizing renewable energy investments at Northeastern University. Currently, the university invests across multiple energy options, including energy efficiency, on-site renewables, off-site procurement, and electrification, without a unified method to systematically compare their financial performance, energy impact, and environmental benefits. This fragmented approach results in inconsistent evaluation, increased uncertainty in decision-making, and the potential for inefficient capital allocation. As a result, the university lacks a reliable mechanism to ensure that investments are optimized to minimize cost, reduce emissions, and achieve long-term sustainability objectives.

### **1.2 Solution**

This project proposes the development of an Integrated Renewable Energy Investment Optimization Model to support data-driven capital allocation decisions at Northeastern University. The model evaluates investment options including energy efficiency measures, on-site renewable energy (solar), off-site renewable procurement (PPAs, RECs), and electrification strategies. It calculates key metrics such as Cost of Saved Energy (CSE), Levelized Cost of Energy (LCOE), payback period, energy savings, and emissions reduction.

The core of the solution is an Excel-based optimization model using Solver algorithms, which determines the optimal allocation of investments across energy alternatives under defined constraints such as budget, demand, and sustainability targets. Python is used for data processing and baseline modeling, while Power BI is used for visualization. This enables a shift from isolated decision-making to a portfolio-level optimization approach, ensuring capital is allocated to the most cost-effective and high-impact investments.

## **2.0 PURPOSE AND OBJECTIVES**

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### **2.1 Purpose**

The purpose of this project is to optimize Northeastern University's energy capital allocation by developing a data-driven investment optimization model that prioritizes energy investments based on financial efficiency, energy impact, and sustainability outcomes. Currently, the university invests across

multiple energy levers, including energy efficiency measures, on-site renewable generation, off-site renewable procurement, and electrification. However, these investments are not evaluated through a unified optimization approach at the portfolio level.

This project addresses that gap by enabling a structured method to minimize Cost of Saved Energy (CSE) and reduce overall portfolio payback period, ensuring that capital is allocated to the most cost-effective and high-impact opportunities. The model supports strategic decision-making by transforming fragmented investment choices into a cohesive, optimized investment strategy, aligned with Northeastern's long-term sustainability goals and carbon neutrality targets.

## **2.2 Objective**

The primary objective of this project is to develop an Integrated Renewable Energy Investment Optimization Model that enables portfolio-level decision-making for campus energy investments.

The sub-objectives of this project are structured to systematically support the development of a robust and data-driven investment optimization framework. First, the project defines clear evaluation criteria by establishing key performance metrics such as cost, technical feasibility, environmental impact, and risk, ensuring that all renewable energy investments are assessed consistently and objectively. Building on this foundation, the project develops an analytical and optimization model that quantitatively evaluates each investment option using financial and operational metrics, including Cost of Saved Energy (CSE), Levelized Cost of Energy (LCOE), payback period, and total energy savings.

## 3.0 TECHNICAL OVERVIEW

### 3.1 Technical Approach

The technical approach for this project is centered on developing an Integrated Renewable Energy Investment Optimization Model that enables Northeastern University to evaluate and prioritize campus energy investments at the portfolio level. The model is designed to support pre-investment decision-making by combining baseline energy analysis, renewable energy estimation, financial evaluation, and optimization into a single structured workflow.

The project uses Python, Excel, Excel Solver, and Power BI as the primary technical resources. Python is used to collect, clean, and structure publicly available electricity demand, tariff, and emissions-factor data. Excel is used as the main analytical platform to build the baseline model, cost and performance calculations, and scenario comparison logic. Excel Solver is used to optimize the allocation of investments across multiple energy alternatives under defined constraints such as budget, energy demand, and sustainability targets. Power BI is used to visualize portfolio performance and communicate the outputs in a decision-friendly format.

The final deliverables of the project include:

- An Excel-based renewable energy investment optimization model
- Scenario analysis outputs comparing alternative investment portfolios
- Visual dashboards showing cost savings, energy performance, payback, and emissions reduction
- A structured decision report presenting the recommended investment strategy

The technical approach is intentionally structured around what Northeastern is actually trying to decide: how to allocate energy capital across competing investment levers to maximize return and reduce payback period. Instead of treating each renewable option as an isolated choice, the model evaluates multiple investment combinations and identifies the most effective overall portfolio.

### 3.2 Framework Architecture

The framework is organized as an integrated analytical pipeline with five connected components:

- 1. Data Collection and Baseline Modeling**

Campus electricity demand, tariff structures, and emissions-related inputs are collected and processed to establish a baseline representation of current energy use and cost performance.

- 2. Renewable Energy Generation Estimation**

Renewable energy output is estimated for relevant investment options using public datasets, technology assumptions, and performance factors.

- 3. Financial and Cost Modeling**

Investment alternatives are evaluated using financial and cost-effectiveness metrics to quantify economic performance.

- 4. Optimization Modeling**

Excel Solver is used to identify the optimal mix of investment alternatives under budgetary, operational, and sustainability constraints.

- 5. Visualization and Decision Support**

Results are summarized through dashboards and structured outputs that allow stakeholders to compare alternatives clearly and make defensible decisions.sssss

This architecture ensures that each part of the model contributes to a single decision-support objective rather than functioning as an isolated analytical exercise. Energy storage and electrification are treated as supporting investment options that improve portfolio performance and operational outcomes, rather than as standalone renewable generation sources.

### 3.3 Analytical Models

The model begins by establishing a campus energy baseline using available electricity consumption data, tariff assumptions, and emissions factors. This baseline represents the current state against which all alternative energy investment portfolios are evaluated. The renewable energy estimation component calculates the expected contribution of selected technologies using performance assumptions and publicly available reference datasets. These estimates provide the technical basis for comparing alternative investments.

The financial model then evaluates each investment option and portfolio combination using the following key metrics:

- Cost of Saved Energy (CSE)
- Levelized Cost of Energy (LCOE)
- Payback Period
- Energy Savings



- Cost Savings
- Emissions Reduction

These metrics are used because they directly reflect the criteria emphasized in the project's purpose, objectives, and presentation narrative. They allow Northeastern to compare investments not only by cost, but also by financial efficiency and sustainability value.

The core of the system is the optimization model, where Excel Solver determines the best allocation of investments across available energy levers. The objective is to minimize Cost of Saved Energy and compress portfolio payback period while satisfying defined constraints such as capital availability, energy demand requirements, and institutional sustainability goals.

### **3.4 Scenario Analysis and Outputs**

Scenario analysis is used to compare multiple investment portfolios against a baseline scenario representing current campus conditions. Each scenario reflects a different combination of investments across energy efficiency, renewable generation, off-site procurement, and electrification.

Sensitivity analysis is performed to evaluate how changes in assumptions such as cost, demand, tariffs, and technology performance affect the results. This helps identify which variables have the greatest influence on investment outcomes and improves the robustness of recommendations.

The model generates the following outputs:

- Optimized investment portfolios
- Comparative scenario tables
- Financial performance summaries
- Projected energy savings
- Emissions reduction estimates
- Dashboard-based visual decision outputs

These outputs provide Northeastern stakeholders with a clear and actionable basis for selecting the most cost-effective and strategically aligned energy investment portfolio.

## 4.0 IMPLEMENTATION PLAN

### 4.1 WBS

The Work Breakdown Structure (WBS) defines the complete scope of activities required to develop the Renewable Energy Investment Advisor framework. Activities are organized hierarchically across five phases: Project Initiation, Project Planning, Project Execution, Project Monitoring, and Project Closure. The structure follows the 8/80 rule, where individual tasks are sized to require no fewer than eight and no more than eighty hours of effort, ensuring sufficient granularity for scheduling and accountability without excessive fragmentation.

The Initiation Phase establishes project authority, scope, and communication structures. The Planning Phase covers literature review, requirements definition, data source identification, and risk planning. The Execution Phase is the most intensive, encompassing data collection, development of the four analytical models, framework integration, scenario analysis, validation, and testing. The Monitoring Phase covers dashboard preparation, structured reporting, and schedule control. The Closure Phase concludes with the final presentation, final report submission, and formal project acceptance. The complete WBS is provided in Appendix A.

### 4.2 Schedule

The project schedule spans from May 5, 2026 to November 4, 2026, covering a total duration of approximately 26 weeks. Task durations and sequencing are derived directly from the WBS, with dependencies defined to reflect logical execution flow and identify the critical path.

The schedule is structured into phases as follows:

**Initiation and Planning (Weeks 1–6):** Stakeholder alignment, requirements definition, and risk planning

**Framework Design and Data Preparation (Weeks 7–11):** Architecture design and data acquisition

**Execution (Weeks 12–20):** Model development, including demand forecasting, generation modeling, financial modeling, and environmental analysis

**Integration and Scenario Analysis (Weeks 21–23):** Model linking, dashboard development, and portfolio evaluation

**Validation, Reporting, and Closure (Weeks 24–26):** Testing, documentation, and final delivery

The Execution phase is the longest and most resource-intensive phase, consistent with the concentration of technical work across model development and integration.

The detailed project schedule is provided in Appendix B.

#### 4.3 Responsibility Chart

A Responsibility Assignment Matrix (RACI) was developed to define the involvement of each project team member and key stakeholder in every task outlined in the WBS. The RACI designations—Responsible, Accountable, Consulted, and Informed—are assigned at the subtask level across all five project phases, ensuring that ownership and decision authority are unambiguous for each activity. The matrix covers eleven distinct roles spanning the core project team and institutional stakeholders.

The RACI structure reveals the relative workload distribution across roles. The Data and Systems Analyst carry the highest number of Responsible assignments across the project lifecycle, reflecting the central role of data acquisition and model input preparation in this analytical framework. The Project Manager holds the highest Accountable count, consistent with overall delivery of ownership. The complete RACI Matrix is provided in Appendix C.

#### 4.4 Resource Allocation

Resource allocation for this project is structured across all roles defined in the stakeholder mapping, spanning four categories: Project Team, Technical and Energy Experts, Institutional Stakeholders, and External Stakeholders. Four team members are currently assigned, each holding one primary role aligned to their professional background and the RACI workload distribution. The remaining roles are budgeted with standard market rates and estimated hours and will be formally recruited prior to project execution. Total hours and costs are estimated across all five project phases, with the Execution Phase accounting for the majority of effort given its concentration of model development, integration, and validation activities. The complete resource allocation plan is provided in Appendix E.

| Team Member          | Role(s)                | Rate / Hour | Total Hours | Total Cost |
|----------------------|------------------------|-------------|-------------|------------|
| Nadukuru, J.S.R.     | Project Manager        | \$50        | 474.2       | \$23,710   |
| Senthilvel, S.       | Sustainability Analyst | \$45        | 205.4       | \$9,243    |
| Sudhandrakumar, S.K. | Data & Systems Analyst | \$55        | 564.5       | \$31,047.5 |

| Team Member | Role(s)           | Rate / Hour | Total Hours   | Total Cost        |
|-------------|-------------------|-------------|---------------|-------------------|
| Lokesh, A.  | Financial Analyst | \$45        | 238.8         | \$10,746          |
|             | <b>Total</b>      |             | <b>1482.9</b> | <b>\$74,746.5</b> |

*Table 4-1: Resource Allocation Plan*

Effort is weighted toward the Execution Phase for all team members, reflecting the concentration of model development, integration, and validation of work in that phase. Senthilvel carries the highest total allocation at 40%, reflecting the combined demands of energy systems analysis and framework architecture across execution. Sudhandrakumar follows at 36%, driven by data pipeline work and validation testing responsibilities. Nadukuru maintains steady effort across all phases as Project Manager, with consistent coordination, reporting, and stakeholder management responsibilities throughout the project lifecycle. Lokesh's allocation peaks in the Execution and Monitoring phases in line with cost-benefit modeling and sustainability reporting activities.

#### 4.5 Stakeholders

Stakeholders for this project encompass the core project team, institutional decision-makers, technical experts, and external parties whose inputs shape the analytical assumptions and whose interests are served by the framework's outputs. The resource allocation plan in Appendix E identifies all roles and their assigned or planned personnel. The following provides a summary of the key stakeholder groups.

The Project Team consists of the four team members whose roles span project management, energy systems analysis, data and systems analysis, financial evaluation, sustainability assessment, framework architecture, and quality assurance. Institutional Stakeholders include the Executive Sponsor, who holds final adoption authority, and the Steering Committee, which provides cross-functional oversight across facilities, finance, sustainability, and IT. Northeastern Facilities Management and the Energy Manager serve as the primary operational client, providing data access and confirming that framework outputs are usable for real investment decisions.

Technical and Energy Experts provide renewable technology expertise, modeling guidance, and building automation context to ensure that analytical assumptions are grounded in engineering reality. External Stakeholders include utility representatives, who supply tariff structures and interval data access pathways, and energy consultants and vendors, whose cost estimates and performance specifications inform the financial modeling. Government and regulatory bodies influence the feasibility of specific configurations through incentive structures and permitting constraints. End users—including facilities operators and the broader campus community—are the ultimate beneficiaries of the framework’s outputs.

## 5.0 EXECUTION PLAN

### 5.1 Project Monitoring

Effective monitoring of the Renewable Energy Investment Advisor framework ensures that all analytical deliverables are completed to a high standard, on schedule, and in alignment with the decision-support objectives established during project initiation. A structured monitoring system will be maintained throughout the project lifecycle, guided by the Work Breakdown Structure (*Appendix A*) and RACI Matrix (*Appendix C*), to track the following key dimensions.

1. **Cost:** All project expenditures — including labor hours logged by each team member, software licensing fees, and data procurement costs — will be tracked against the approved budget on a biweekly basis. Variance thresholds will be established to trigger escalation if actual costs deviate more than 10% from planned values at any phase boundary.
2. **Schedule:** Task start and completion dates will be compared against the project schedule (*Appendix B*) at every biweekly review. Key milestones — including stakeholder sign-off on framework requirements, completion of each analytical model, integration of the scenario dashboard, and delivery of the final decision report — will be monitored as schedule control points. Any slippage identified in the Execution Phase will be evaluated for downstream impact on the Monitoring and Closure phases, given the sequential dependency between model validation and reporting activities.
3. **Resources:** Resource utilization will be tracked across all four team members in alignment with the RACI Matrix assignments (*Appendix C*). The Project Manager will monitor whether individual workloads remain consistent with planned hour allocations detailed in the Resource Allocation Plan (*Appendix E*), and flag any imbalance that could affect the quality or timeliness of model outputs. Availability of external stakeholders — particularly the Facilities and Energy Manager, who is the primary data provider — will also be tracked, as delays in data access directly affect the Execution Phase schedule.
4. **Quality:** Each analytical model will be subject to defined quality checkpoints prior to integration. These include cross-validation of model outputs against reference datasets, independent review of formula logic and input traceability, and usability assessments to confirm that non-technical

stakeholders can interpret the scenario comparison dashboard. Quality standards will be established during the Planning Phase and enforced throughout execution.

5. **Scope and Changes:** A change control process will be maintained to evaluate any requests that expand the framework beyond its four defined model components or extend scenario analysis beyond the planned configurations. All change requests will be assessed for schedule, cost, and quality impact before approval by the Project Manager and Executive Sponsor.
6. **Risks:** Active risks identified in Section 6.0 will be reviewed at each biweekly meeting, with risk owners providing status updates on mitigation activities and the risk register updated to reflect any changes in likelihood or impact.
7. **Key Performance Indicators:** The following KPIs will be tracked throughout the project lifecycle to provide an objective, quantitative basis for assessing project health at each biweekly review.

|                                                                                                                                                   |                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Schedule performance</b><br>Phase-level Schedule Performance Index (SPI), target $\geq 0.90$                                                   | <b>Cost performance</b><br>Phase-level Cost Performance Index (CPI), target $\geq 0.90$                                                                |
| <b>Data collection progress</b><br>% of planned data acquisition tasks completed on schedule, target $\geq 85\%$ by end of Data Acquisition phase | <b>Model validation pass rate</b><br>% of model outputs passing first-pass validation review without rework, target $\geq 80\%$                        |
| <b>Stakeholder review resolution</b><br>Open stakeholder feedback items resolved within 5 business days of receipt, target 100%                   | <b>Contingency consumption</b><br>Cumulative contingency usage as % of reserve; escalation threshold at 50% consumed before Execution Phase completion |

Biweekly project status meetings will serve as the primary forum for reviewing monitoring data, coordinating across workstreams, and initiating corrective action. All four team members will participate, and a written status summary will be distributed after each session.

## 5.2 Project Control

Project control will be implemented through a structured feedback system in which each functional area maintains ownership of its respective control domain. Biweekly status meetings serve as the primary control mechanism, with each team member reporting on progress, flagging deviations, and proposing corrective actions within their area of responsibility. Control responsibilities are assigned as follows.

| Key Factor                                         | Role Responsible                              |
|----------------------------------------------------|-----------------------------------------------|
| Cost and budget variance                           | Financial Analyst (Lokesh, A.)                |
| Schedule and milestone adherence                   | Project Manager (Nadukuru, J.S.R.)            |
| Model quality and technical accuracy               | Data & Systems Analyst (Sudhandrakumar, S.K.) |
| Environmental outputs and sustainability alignment | Sustainability Analyst (Senthilvel, S.)       |
| Scope changes and stakeholder sign-offs            | Project Manager (Nadukuru, J.S.R.)            |

*Table 5-1: Project Control Responsibility Matrix*

The Project Manager holds overall delivery accountability and will coordinate corrective actions when deviations span multiple control domains. Where a corrective action affects cost, schedule, or scope simultaneously, the full project team will convene to assess trade-offs before implementation. All approved changes will be documented and reflected in updated versions of the project schedule and budget.

An earned value approach will be used to assess schedule and cost performance at phase boundaries. Planned value, earned value, and actual cost will be compared at the conclusion of each phase to generate Schedule Performance Index and Cost Performance Index values. These metrics will inform resource reallocation decisions entering subsequent phases, particularly at the transition into the Execution Phase, where the majority of labor effort is concentrated (*Appendix D, Appendix E*).

### **5.3 Project Audit**

A structured audit process will be conducted at two points: a mid-project review at the conclusion of the Framework Design and Data Acquisition phases, and a final audit prior to project closure. The purpose of these audits is to verify that the project is being executed in accordance with the approved plan, that analytical outputs are traceable and defensible, and that deliverables meet the quality standards required for institutional stakeholder review.

1. **Analytical model audit:** Each of the four model components — energy demand forecasting, renewable production estimation, cost-benefit analysis, and environmental impact assessment — will be independently reviewed for formula accuracy, assumption documentation, and internal consistency. Outputs will be cross-checked against reference datasets and industry benchmarks. The Data and Systems Analyst will lead this review, with validation support from the Renewable Energy Engineer where generation assumptions are involved.
2. **Documentation and traceability audit:** All input assumptions, data sources, and calculation logic embedded in the Excel framework will be reviewed to confirm they are clearly documented and auditable. Given the institutional context of this project, the framework must be interpretable by Northeastern staff who were not involved in its development. The audit will verify that each input can be traced to a documented source and that the logic connecting inputs to outputs is transparent and reproducible.
3. **Financial audit:** Actual expenditures will be compared to the approved budget (*Appendix D*) at each audit point. All labor hours logged and external costs incurred will be reviewed for accuracy and appropriate authorization. Any use of contingency funds will be documented with justification.
4. **Stakeholder alignment audit:** At the mid-project review, the team will conduct a structured check-in with the Facilities and Energy Manager and Executive Sponsor to confirm that framework design and scenario definitions remain aligned with institutional decision-making needs. Feedback gathered will be documented and assessed for scope impact before model integration activities begin.

Audit findings will be compiled into a written report following each review, distributed to the project team and sponsor, and used to inform corrective actions before the next phase begins.

#### 5.4 Project Closure

The Project Closure Phase formalizes the completion of all deliverables, ensures a structured handover to the client, and captures lessons learned to support future analytical work at Northeastern. Closure activities are organized across five areas.

1. **Deliverable review and acceptance:** All project deliverables — including the integrated Excel-based framework, scenario comparison dashboard, and structured decision report — will be reviewed against the acceptance criteria established during project initiation. The Project Manager will coordinate final review sessions with the Facilities and Energy Manager and Executive Sponsor, and formal written acceptance will be obtained before the project is considered closed.



2. **Documentation and handover:** A comprehensive user guide will be prepared to enable Northeastern staff to operate and update the framework independently. This guide will document input data requirements, assumption update procedures, scenario configuration instructions, and output interpretation guidance. All project documentation — including the final decision report, model files, data sources, and assumption logs — will be compiled and archived in a format accessible to the Facilities and Energy Management Department for ongoing institutional reference.
3. **Financial closure:** All project expenditures will be reconciled against the approved budget (*Appendix D*). A final budget report comparing planned versus actual costs will be prepared, documenting any variances and explaining the use of contingency reserves, and submitted to the Executive Sponsor as part of the formal sign-off package.
4. **Team dissolution and recognition:** Following formal project acceptance, team members will be released from their project assignments. Exit discussions will be conducted to gather individual reflections on project execution, team dynamics, and technical challenges encountered. Contributions of all team members will be formally acknowledged in the project closure report.
5. **Post-implementation review:** A post-implementation review will be recommended to the Facilities and Energy Management Department approximately three to six months after framework delivery. This review will assess whether the framework has been actively used in investment planning, whether assumption or scenario updates have been required, and whether the decision report insights have influenced capital planning outcomes. Findings will contribute to institutional knowledge around renewable energy investment planning and inform any future iterations of the framework.

## 6.0 RISK ASSESSMENT MANAGEMENT PLAN

### 6.1 Identification and Analysis of Risks

This section identifies the principal risks associated with the development and delivery of the Renewable Energy Investment Advisor framework, evaluates their likelihood and potential impact on project outcomes, and establishes the basis for the mitigation strategies detailed in Section 6.2. Risks are grouped into four categories: Data and Information, Stakeholder and Scope, Technical, and Schedule and Financial.

**Data and information risks:** The framework’s analytical outputs depend on the availability of accurate, site-specific data from Northeastern — including historical campus energy consumption records, utility tariff structures, and renewable resource datasets. Delays or gaps in data access from the Facilities and Energy Management Department represent the highest-likelihood risk to the Execution Phase schedule. Incomplete or inconsistent data could also undermine the credibility of model outputs and reduce stakeholder confidence in framework results.

**Stakeholder and scope risks:** Several key inputs — including decision criteria, scenario assumptions, and validation feedback — require active participation from institutional stakeholders. If stakeholder availability is limited during critical milestones, schedule delays may propagate across downstream phases. Additionally, given the breadth of institutional interest in sustainability planning, there is a risk that expectations expand beyond the defined scope — for example, into implementation planning or procurement activities — significantly increasing project complexity.

**Technical risks:** The Excel-based environment imposes inherent constraints on model complexity and computational capacity. If scenario analysis requirements exceed what can be reliably implemented within Excel, the team may face trade-offs between analytical depth and platform feasibility. Separately, errors in formula logic or misalignment between model assumptions and real-world conditions could produce misleading outputs that reduce the framework’s defensibility for institutional decision-making.

**Schedule and financial risks:** The Execution Phase concentrates the majority of model development, integration, and validation activities within a nine-week window. Parallel workstreams during this period create schedule risk if dependencies between model components are not resolved in the planned sequence. Budget overruns may result from underestimated effort in complex model development tasks or additional data procurement costs. The complete risk register with likelihood, impact, and priority ratings is summarized in Table 6-1, and risks are visualized across the probability-impact matrix in Figure 6-1.

| #  | Risk                                              | Category  | Likelihood | Impact | Priority |
|----|---------------------------------------------------|-----------|------------|--------|----------|
| R1 | Delayed access to campus energy data              | Data      | High       | High   | Critical |
| R2 | Incomplete or inconsistent data quality           | Data      | Medium     | High   | High     |
| R3 | Scope expansion beyond defined framework boundary | Scope     | Medium     | High   | High     |
| R4 | Formula or assumption errors in model outputs     | Technical | Medium     | High   | High     |

| #  | Risk                                              | Category    | Likelihood | Impact | Priority |
|----|---------------------------------------------------|-------------|------------|--------|----------|
| R5 | Stakeholder unavailability at sign-off milestones | Stakeholder | Medium     | Medium | Moderate |
| R6 | Execution Phase schedule slippage                 | Schedule    | Medium     | Medium | Moderate |
| R7 | Excel platform limitations for complex modeling   | Technical   | Low        | Medium | Moderate |
| R8 | Budget overrun due to effort underestimation      | Financial   | Low        | Medium | Low      |

Table 6-1: Risk Register

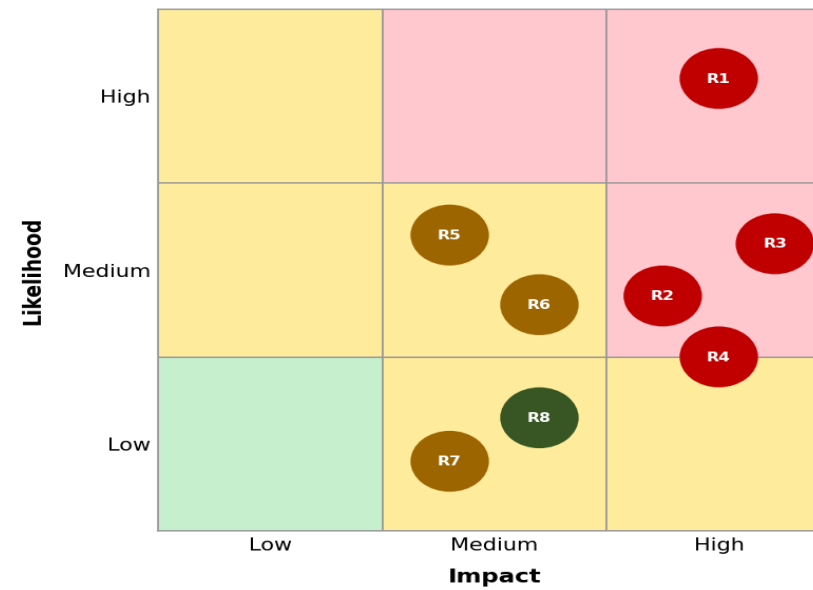


Figure 6-1: Probability-Impact Risk Matrix

## 6.2 Risk Management Plan

The risk management plan establishes a mitigation strategy, responsible owner, and escalation trigger for each risk identified in Section 6.1. Strategies are designed to reduce either the likelihood of occurrence, the impact if the risk materializes, or both. The complete plan is summarized in Table 6-2.

| #  | Mitigation Strategy                                                                                                                               | Owner                  | Contingency Action                                                                                                                  | Escalation Trigger                                                                           |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| R1 | Formalize data access agreements with Facilities Management during Planning Phase; develop data readiness checklist before Execution begins.      | Data & Systems Analyst | Substitute proxy datasets and publicly available benchmarks; document all substitutions as flagged assumptions in framework inputs. | Any required dataset unavailable 2 weeks before Execution Phase start.                       |
| R2 | Define data quality acceptance criteria upfront; validate all collected datasets against criteria before model input preparation.                 | Data & Systems Analyst | Apply data cleaning and imputation procedures; document all adjustments with justification in assumption log.                       | More than 20% of input records fail quality validation.                                      |
| R3 | Document and formally communicate scope boundary to all stakeholders at requirements sign-off; enforce change control for all expansion requests. | Project Manager        | Conduct formal scope impact assessment; present trade-offs to Executive Sponsor for approval before any change is implemented.      | Any request requiring new model components or technology configurations not in original WBS. |
| R4 | Require independent second-reviewer sign-off on all model formulas before                                                                         | Data & Systems Analyst | Isolate affected model component; re-run validation cycle; delay integration until                                                  | Model validation pass rate falls below 80% threshold                                         |

| #         | Mitigation Strategy                                                                                                                                        | Owner                  | Contingency Action                                                                                                                     | Escalation Trigger                                                                                          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
|           | integration; validate outputs against reference datasets.                                                                                                  |                        | rework is complete.                                                                                                                    | (see Section 5.1 KPIs).                                                                                     |
| <b>R5</b> | Establish stakeholder engagement calendar during initiation; distribute review materials 5 business days before each milestone meeting.                    | Project Manager        | Collect feedback via annotated documents asynchronously; set a 48-hour response deadline to avoid blocking schedule.                   | Sign-off not received within 5 business days of scheduled milestone.                                        |
| <b>R6</b> | Map all Execution Phase task dependencies explicitly; incorporate two-week schedule buffer at Execution–Monitoring phase boundary.                         | Project Manager        | Initiate scope prioritization review to identify deferrable activities; reallocate hours from lower-priority tasks.                    | SPI falls below 0.90 at any phase-boundary review.                                                          |
| <b>R7</b> | Prototype scenario comparison and sensitivity analysis modules during Framework Design phase to validate Excel feasibility before full development begins. | Data & Systems Analyst | Reduce scenario complexity or restructure calculation logic; present revised approach to Executive Sponsor.                            | Prototype reveals performance or accuracy degradation under planned scenario load.                          |
| <b>R8</b> | Track labor hours and expenditures biweekly against phase-level budget; review CPI at every phase boundary.                                                | Financial Analyst      | Reallocate effort from lower-priority tasks; request Executive Sponsor authorization before drawing on contingency reserve beyond 50%. | CPI falls below 0.90, or cumulative spend exceeds 90% of approved budget before Execution Phase completion. |

*Table 6-2: Risk Management Plan*

### **6.3 Continuous Risk Assessment**

Risk assessment will be treated as an ongoing process rather than a one-time planning activity. The risk register established during the Planning Phase will serve as a living document, updated at each biweekly project status meeting to reflect changes in the likelihood or impact of identified risks, the emergence of new risks, and the effectiveness of active mitigation strategies. All updates will be version-controlled and maintained alongside the project schedule and budget documentation (*Appendix B, Appendix D*).

Each team member will serve as a risk monitor within their functional domain. The Data and Systems Analyst will flag data quality or model stability issues as they arise during development. The Financial Analyst will track budget variance trends and identify early signals of cost overrun risk. The Sustainability Analyst will monitor changes in Northeastern’s sustainability reporting requirements or institutional priorities that could affect framework scope or output alignment. The Project Manager will consolidate risk updates, maintain the risk register, and determine when escalation to the Executive Sponsor is warranted.

The KPIs defined in Section 5.1 — including phase-level SPI and CPI values, data collection completion rates, and model validation pass rates — will serve as the primary early-warning indicators between formal review meetings. When any indicator crosses its predefined threshold, the responsible team member will initiate an out-of-cycle risk review rather than waiting for the next scheduled meeting.

Lessons learned from risk events encountered during the project will be documented in the project closure report and made available to the Facilities and Energy Management Department as part of the framework handover. This documentation will support institutional knowledge for future renewable energy planning initiatives and improve risk preparedness for any subsequent iterations of the analytical framework.

## **7.0 FINANCIAL PLAN WITH BUDGET**

### **7.1 High-Level Details**

The financial plan for the Renewable Energy Investment Advisor framework establishes the total cost of developing, validating, and delivering a structured decision-support tool for renewable energy investment for Northeastern. The budget is organized across five project phases and covers all labor, materials, and miscellaneous costs required for full project execution. A 7% contingency is included to account for scope adjustments, data access delays, or unforeseen technical challenges. The complete task-level budget breakdown is provided in Appendix D.

| Budget Category       | Amount              | % of Total   |
|-----------------------|---------------------|--------------|
| Category              | Amount              | % of Total   |
| Labor                 | \$146,350           | 93.6%        |
| Materials & Equipment |                     | 0.3%         |
| Miscellaneous         |                     | 0.1%         |
| <b>Subtotal</b>       | <b>\$146,350</b>    | <b>94.0%</b> |
| 7% Contingency        | \$10,244.50         | 6.0%         |
| <b>Grand Total</b>    | <b>\$156,594.50</b> | <b>100%</b>  |

*Table 7-1: Budget Summary*

Labor represents the dominant cost category at 93.6% of the total budget, reflecting the analytical and knowledge-intensive nature of this project. The framework is built entirely through expert effort — data collection, model development, validation, and reporting — with minimal physical materials required. Materials and equipment costs are limited to software licensing and data access tools. The 7% contingency of \$3,364,661.93 provides a buffer for schedule adjustments, additional data procurement, or scope refinements identified during execution.

By phase, the Execution Phase carries the largest budget share, accounting for approximately 52% of total labor cost. This reflects the concentration of model development, integration, scenario analysis, and validation activities in that phase. The Initiation Phase is the lightest, representing roughly 6% of total labor, consistent with its focus on alignment and planning activities prior to technical work.

## 7.2 Budget Justification

The budget for the Renewable Energy Investment Advisor framework is justified on the basis that structured, evidence-based decision-making produces measurably better investment outcomes than ad hoc approaches. Northeastern facilities routinely commit capital expenditures in the range of \$1,000,000 to \$5,000,000 for renewable energy projects. A decision support framework that costs \$3,600,188.27 to develop and improves the quality of those decisions — by reducing the probability of selecting a suboptimal technology configuration or misallocating capital — generates a return that substantially exceeds its development cost.





## 8.0 TEAM CREDENTIALS

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### **Nadukuru, Jyoti Swarup Rao**

N J Swarup Rao is a master's student at Northeastern University majoring in Engineering Management with comprehensive skills in data-driven decision making, operations management, and coordination and management of multi-functional programs. Before his postgraduate course, he worked as a Senior Operations Associate for a maritime services firm and efficaciously managed and uplifted workflow efficiency and operational cost reduction by strategically negotiating and determining contracts. Currently, he also possesses pertinent skills in undertaking statistical data analysis for large public data sets and assistive wearable device programming for sign-to-speech and text translation. In addition, he has expertise and experience in programming languages and project management tools and is efficacious in bridging the gap between engineering execution and management decision making, making him a potential and skilled individual for assuming leadership positions for scope planning and management, optimization, and performance control activities.

### **Senthilvel, Sujeeth**

Sujeeth is currently a master's student in Engineering Management with a strong foundation in systems engineering, simulation-driven decision making, and cross-functional program execution. Prior to graduate studies, Sujeeth worked in engineering and product-facing roles supporting large-scale industrial programs for Fortune 100 organizations, where he contributed to simulation workflows, systems validation, and operational decision making across cross-functional teams. His expertise in integrating technical depth with managerial judgement positions him to lead scope definition, schedule control, and performance control, ensuring that project decisions are technically sound and operationally viable.

### **Sudhandrakumar, Santhosh Kumar**

Santhosh is a graduate student at Northeastern University pursuing a Master's degree in Engineering Management. He brings hands-on industry experience from Tata Consultancy Services (TCS), where he worked as a Manual Tester and collaborated closely with clients to understand requirements, demonstrate product features, and communicate feedback to development teams. Through this role, he developed strong skills in requirements analysis, stakeholder communication, and cross-functional coordination. His blend of technical knowledge and client-facing experience will help ensure clear communication and efficient execution as the project progresses.

### **Lokesh, Ayush**

Ayush Lokesh is a master's student pursuing a degree in Engineering Management, Northeastern University, which has a significant technical component in data analysis, system modeling, and decision support techniques. Ayush also has industry experience, as he interned at Inceptbio, which is a startup

environment, and here he contributed by working with data analysis, problem solving techniques, and the evaluation of complex systems within real world constraints. For this project, Ayush Lokesh can contribute in terms of developing analytical frameworks, while also undertaking financial and risk scenario analysis, as well as data driven evaluation of renewable energy investment alternatives.

## APPENDICES

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### APPENDIX A: Work Breakdown Structure

Table A-1: Work Breakdown Structure

| ID           | TASKS                                                                 |
|--------------|-----------------------------------------------------------------------|
| <b>1.0</b>   | <b>Project Initiation</b>                                             |
| <b>1.1</b>   | Conduct project kick-off meeting                                      |
| <b>1.1.1</b> | Define project scope, boundaries, and success criteria                |
| <b>1.1.2</b> | Identify and map stakeholders                                         |
| <b>1.2</b>   | Develop project charter                                               |
| <b>1.3</b>   | Establish communication plan and reporting cadence                    |
| <b>1.4</b>   | Obtain project sponsor approval and sign-off                          |
|              |                                                                       |
| <b>2.0</b>   | <b>Project Planning</b>                                               |
| <b>2.1</b>   | Requirements Definition                                               |
| <b>2.1.1</b> | Conduct stakeholder interviews to gather decision-making requirements |

|       |                                                                   |
|-------|-------------------------------------------------------------------|
| 2.1.2 | Define evaluation criteria and framework objectives               |
| 2.1.3 | Document initial assumptions, constraints, and exclusions         |
| 2.1.4 | Obtain stakeholder sign-off on requirements                       |
| 2.2   | Research and Background Analysis                                  |
| 2.2.1 | Review literature on renewable energy technologies                |
| 2.2.2 | Identify and catalog data sources and regulatory incentives       |
| 2.3   | Project Management Planning                                       |
| 2.3.1 | Develop detailed project schedule and milestones                  |
| 2.3.2 | Assign roles and responsibilities (RACI)                          |
| 2.3.3 | Develop risk management plan                                      |
| 2.3.4 | Develop quality management plan                                   |
|       |                                                                   |
| 3.0   | <b>Framework Design</b>                                           |
| 3.1   | Design framework architecture and data flow between components    |
| 3.2   | Establish input/output specifications for each model              |
| 3.3   | Design scenario comparison and sensitivity analysis structure     |
| 3.4   | Define Excel workbook structure, layout, and data entry standards |
| 3.5   | Obtain stakeholder review and approval of framework design        |
|       |                                                                   |
| 4.0   | <b>Data Acquisition and Preparation</b>                           |

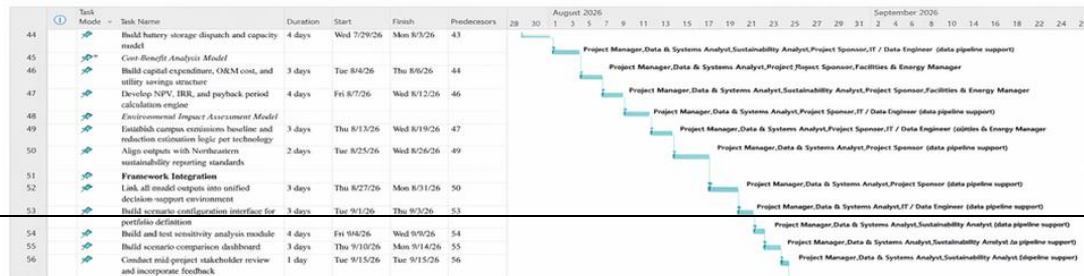
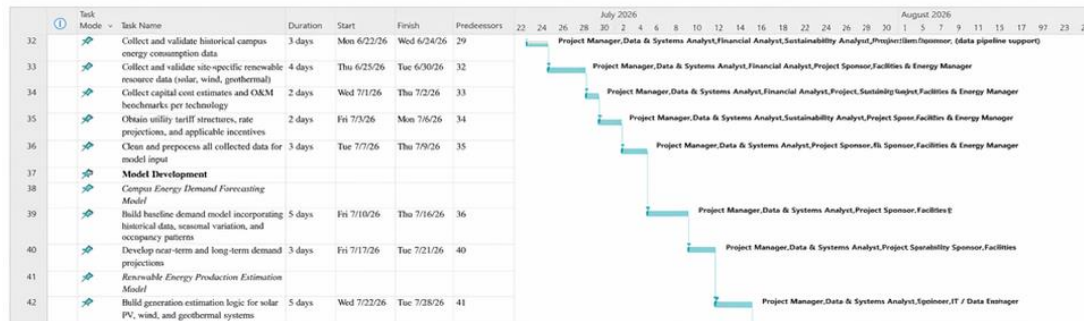
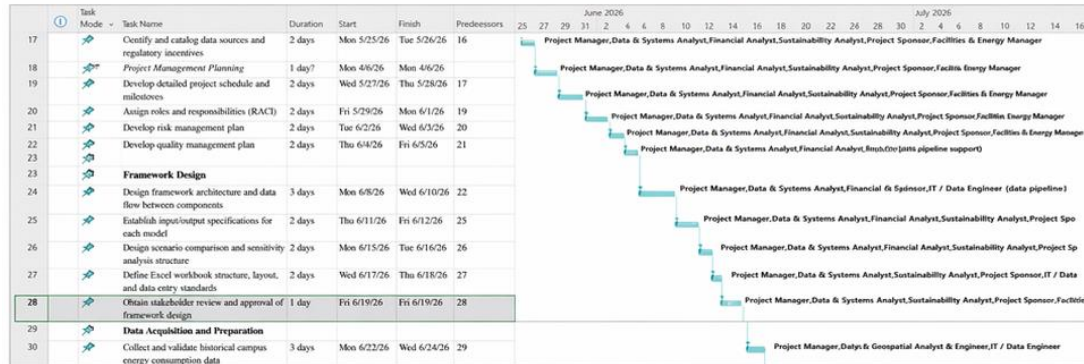
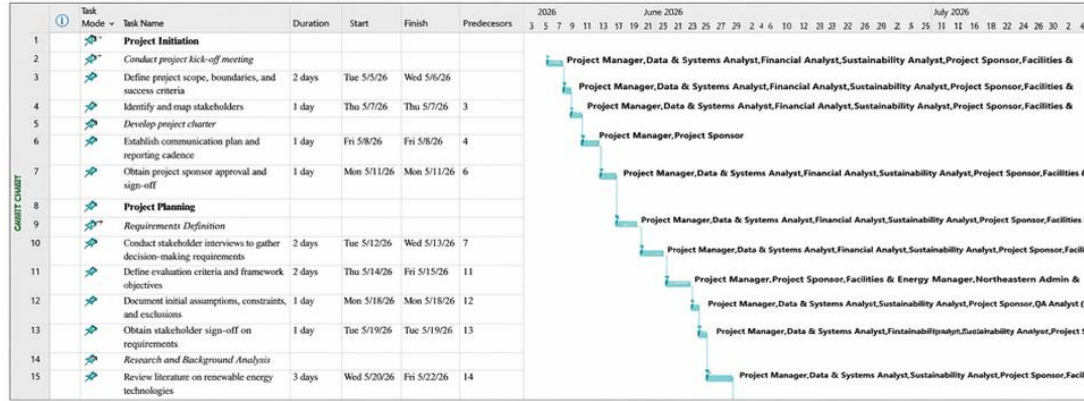
|       |                                                                                                       |
|-------|-------------------------------------------------------------------------------------------------------|
| 4.1   | Collect and validate historical campus energy consumption data                                        |
| 4.2   | Collect and validate site-specific renewable resource data (solar, wind, geothermal)                  |
| 4.3   | Collect capital cost estimates and O&M benchmarks per technology                                      |
| 4.4   | Obtain utility tariff structures, rate projections, and applicable incentives                         |
| 4.5   | Clean and preprocess all collected data for model input                                               |
|       |                                                                                                       |
| 5.0   | <b>Model Development</b>                                                                              |
| 5.1   | Campus Energy Demand Forecasting Model                                                                |
| 5.1.1 | Build baseline demand model incorporating historical data, seasonal variation, and occupancy patterns |
| 5.1.2 | Develop near-term and long-term demand projections                                                    |
| 5.2   | Renewable Energy Production Estimation Model                                                          |
| 5.2.1 | Build generation estimation logic for solar PV, wind, and geothermal systems                          |
| 5.2.2 | Build battery storage dispatch and capacity model                                                     |
| 5.3   | Cost-Benefit Analysis Model                                                                           |
| 5.3.1 | Build capital expenditure, O&M cost, and utility savings structure                                    |
| 5.3.2 | Develop NPV, IRR, and payback period calculation engine                                               |
| 5.4   | Environmental Impact Assessment Model                                                                 |
| 5.4.1 | Establish campus emissions baseline and reduction estimation logic per technology                     |
| 5.4.2 | Align outputs with Northeastern sustainability reporting standards                                    |
|       |                                                                                                       |

|            |                                                                                   |
|------------|-----------------------------------------------------------------------------------|
| <b>6.0</b> | <b>Framework Integration</b>                                                      |
| <b>6.1</b> | Link all model outputs into unified decision-support environment                  |
| <b>6.2</b> | Build scenario configuration interface for portfolio definition                   |
| <b>6.3</b> | Build and test sensitivity analysis module                                        |
| <b>6.4</b> | Build scenario comparison dashboard                                               |
| <b>6.5</b> | Conduct mid-project stakeholder review and incorporate feedback                   |
|            |                                                                                   |
| <b>7.0</b> | <b>Scenario Analysis</b>                                                          |
| <b>7.1</b> | Define and run baseline scenario reflecting current campus energy conditions      |
| <b>7.2</b> | Define and run alternative investment scenarios across technology combinations    |
| <b>7.3</b> | Execute sensitivity analysis on critical input assumptions                        |
| <b>7.4</b> | Compile internal working documentation of scenario results and trade-off findings |
|            |                                                                                   |
| <b>8.0</b> | <b>Validation and Quality Assurance</b>                                           |
| <b>8.1</b> | Validate all model outputs against reference datasets and industry benchmarks     |
| <b>8.2</b> | Test integrated framework for edge cases and boundary conditions                  |
| <b>8.3</b> | Review framework usability for non-technical stakeholders                         |
| <b>8.4</b> | Audit input documentation and assumption traceability                             |
| <b>8.5</b> | Address validation findings and conduct final internal review                     |
|            |                                                                                   |

|             |                                                                        |
|-------------|------------------------------------------------------------------------|
| <b>9.0</b>  | <b>Reporting and Delivery</b>                                          |
| <b>9.1</b>  | Write user guide and finalize methodology documentation                |
| <b>9.2</b>  | Prepare scenario comparison summary tables and visualizations          |
| <b>9.3</b>  | Synthesize findings into structured client-facing decision report      |
| <b>9.4</b>  | Draft investment strategy recommendations aligned to campus objectives |
| <b>9.5</b>  | Incorporate final stakeholder feedback into decision report            |
|             |                                                                        |
| <b>10.0</b> | <b>Project Closure</b>                                                 |
| <b>10.1</b> | Prepare and rehearse final project presentation                        |
| <b>10.2</b> | Submit final report and obtain formal project acceptance from sponsor  |

## APPENDIX B: Project Schedule

Table B-1: Project Schedule (May 5 – November 4 , 2026)



## APPENDIX C: RACI Matrix

**Table C-1: RACI Responsibility Assignment Matrix**

| Task / Deliverables                                          | Project Manager | Data & Systems Analyst | Financial Analyst | Sustainability Analyst | Project Sponsor | Facilities & Energy Manager | IT / Data Engineer | Renewable Energy Engineer | Energy Consultant | QA Analyst | Northeastern Admin & Energy SPOC |
|--------------------------------------------------------------|-----------------|------------------------|-------------------|------------------------|-----------------|-----------------------------|--------------------|---------------------------|-------------------|------------|----------------------------------|
| <b>1.0 Project Initiation</b>                                |                 |                        |                   |                        |                 |                             |                    |                           |                   |            |                                  |
| 1.1 Conduct project kick-off meeting                         |                 |                        |                   |                        |                 |                             |                    |                           |                   |            |                                  |
| 1.1.1 Define project scope, boundaries, and success criteria | R               | C                      | C                 | C                      | A               | C                           |                    |                           |                   |            | I                                |
| 1.1.2 Identify and map stakeholders                          | R               | C                      | C                 | C                      | A               | C                           | C                  | C                         | C                 |            | I                                |
| 1.2 Develop project charter                                  | R               |                        |                   | C                      | A               | C                           |                    |                           |                   |            |                                  |
| 1.3 Establish communication plan and reporting cadence       | R               | I                      | I                 | I                      | A               | I                           | I                  | I                         |                   |            | C                                |
| 1.4 Obtain project                                           | R               |                        |                   |                        | A               |                             |                    |                           |                   |            |                                  |



|                                                                             |   |   |   |   |   |   |  |   |   |  |   |
|-----------------------------------------------------------------------------|---|---|---|---|---|---|--|---|---|--|---|
| sponsor approval and sign-off                                               |   |   |   |   |   |   |  |   |   |  |   |
|                                                                             |   |   |   |   |   |   |  |   |   |  |   |
| <b>2.0 Project Planning</b>                                                 |   |   |   |   |   |   |  |   |   |  |   |
| 2.1 Requirements Definition                                                 |   |   |   |   |   |   |  |   |   |  |   |
| 2.1.1 Conduct stakeholder interviews to gather decision-making requirements | A | C | C | C | I | R |  | C |   |  |   |
| 2.1.2 Define evaluation criteria and framework objectives                   | A | R | C | C | I | C |  | C | C |  |   |
| 2.1.3 Document initial assumptions, constraints, and exclusions             | R | A | C | C | I | C |  |   |   |  |   |
| 2.1.4 Obtain stakeholder sign-off on requirements                           | R |   |   |   | A | C |  |   |   |  | I |
| 2.2 Research and Background Analysis                                        |   |   |   |   |   |   |  |   |   |  |   |
| 2.2.1 Review literature on                                                  | I | C |   | R | I |   |  | A | C |  |   |

|                                                                   |   |   |   |   |   |   |   |   |   |   |  |
|-------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|--|
| renewable energy technologies                                     |   |   |   |   |   |   |   |   |   |   |  |
| 2.2.2 Identify and catalog data sources and regulatory incentives | I | A | C | C | I | C | R | C | C |   |  |
| 2.3 Project Management Planning                                   |   |   |   |   |   |   |   |   |   |   |  |
| 2.3.1 Develop detailed project schedule and milestones            | R | C | C | C | A | C |   |   |   |   |  |
| 2.3.2 Assign roles and responsibilities (RACI)                    | R | C | C | C | A |   |   |   |   |   |  |
| 2.3.3 Develop risk management plan                                | R | C | C | C | A | C |   | C |   |   |  |
| 2.3.4 Develop quality management plan                             | R | C |   | C | A |   |   |   |   | A |  |
|                                                                   |   |   |   |   |   |   |   |   |   |   |  |
| <b>3.0 Framework Design</b>                                       |   |   |   |   |   |   |   |   |   |   |  |
| 3.1 Design framework architecture and                             | A | R |   |   | I |   | C | C |   |   |  |

|                                                                       |   |   |   |   |   |   |   |   |   |  |   |
|-----------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|--|---|
| data flow between components                                          |   |   |   |   |   |   |   |   |   |  |   |
| 3.2 Establish input/output specifications for each model              | A | R | C | C | I | C | C | C |   |  |   |
| 3.3 Design scenario comparison and sensitivity analysis structure     | A | R | C | C | I |   |   | C |   |  |   |
| 3.4 Define Excel workbook structure, layout, and data entry standards | A | R |   |   | I |   | C |   |   |  |   |
| 3.5 Obtain stakeholder review and approval of framework design        | R | C | C | C | A | C |   |   |   |  | I |
|                                                                       |   |   |   |   |   |   |   |   |   |  |   |
| <b>4.0 Data Acquisition and Preparation</b>                           |   |   |   |   |   |   |   |   |   |  |   |
| 4.1 Collect and validate historical campus energy consumption data    | I | A |   |   | I | R | C |   |   |  |   |
| 4.2 Collect and validate site-                                        | I | A |   | C | I | C | C | R | C |  |   |

|                                                                                                    |   |   |   |   |   |   |   |   |   |  |  |
|----------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|--|--|
| specific renewable resource data (solar, wind, geothermal)                                         |   |   |   |   |   |   |   |   |   |  |  |
| 4.3 Collect capital cost estimates and O&M benchmarks per technology                               | I | C | A |   | I | C |   | R | R |  |  |
| 4.4 Obtain utility tariff structures, rate projections, and applicable incentives                  | I | C | A |   | I | C |   |   | R |  |  |
| 4.5 Clean and preprocess all collected data for model input                                        | I | R |   |   | I |   | A |   |   |  |  |
|                                                                                                    |   |   |   |   |   |   |   |   |   |  |  |
| <b>5.0 Model Development</b>                                                                       |   |   |   |   |   |   |   |   |   |  |  |
| 5.1 Campus Energy Demand Forecasting Model                                                         |   |   |   |   |   |   |   |   |   |  |  |
| 5.1.1 Build baseline demand model incorporating historical data, seasonal variation, and occupancy | I | R |   | C | I | C | A | C |   |  |  |

|                                                                                    |   |   |   |   |   |   |   |   |   |  |  |
|------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|--|--|
| patterns                                                                           |   |   |   |   |   |   |   |   |   |  |  |
| 5.1.2 Develop near-term and long-term demand projections                           | I | R | C | C | I | C |   | C |   |  |  |
| 5.2 Renewable Energy Production Estimation Model                                   |   |   |   |   |   |   |   |   |   |  |  |
| 5.2.1 Build generation estimation logic for solar PV, wind, and geothermal systems | I | A |   | C | I |   | C | R | C |  |  |
| 5.2.2 Build battery storage dispatch and capacity model                            | I | A |   | C | I |   | C | R | C |  |  |
| 5.3 Cost-Benefit Analysis Model                                                    |   |   |   |   |   |   |   |   |   |  |  |
| 5.3.1 Build capital expenditure, O&M cost, and utility savings structure           | I | C | R |   | I |   |   | C | A |  |  |
| 5.3.2 Develop NPV, IRR, and payback period calculation engine                      | I | C | R |   | I |   |   | C | C |  |  |
| 5.4 Environmental Impact Assessment Model                                          |   |   |   |   |   |   |   |   |   |  |  |

|                                                                                         |   |   |   |   |   |   |   |   |  |  |   |
|-----------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|--|--|---|
| 5.4.1 Establish campus emissions baseline and reduction estimation logic per technology | I | C |   | R | I | C |   | C |  |  |   |
| 5.4.2 Align outputs with Northeastern sustainability reporting standards                | I |   |   | R | A | C |   |   |  |  |   |
|                                                                                         |   |   |   |   |   |   |   |   |  |  |   |
| <b>6.0 Framework Integration</b>                                                        |   |   |   |   |   |   |   |   |  |  |   |
| 6.1 Link all model outputs into unified decision-support environment                    | I | R | C | C | I |   | A | C |  |  |   |
| 6.2 Build scenario configuration interface for portfolio definition                     | I | R |   |   | I |   | A |   |  |  |   |
| 6.3 Build and test sensitivity analysis module                                          | I | R | C |   | I |   | A | C |  |  |   |
| 6.4 Build scenario comparison dashboard                                                 | A | R | C | C | I | C | C |   |  |  |   |
| 6.5 Conduct mid-project stakeholder                                                     | R | C | C | C | A | C |   |   |  |  | I |

|                                                                                       |   |   |   |   |   |   |  |   |   |  |  |
|---------------------------------------------------------------------------------------|---|---|---|---|---|---|--|---|---|--|--|
| review and incorporate feedback                                                       |   |   |   |   |   |   |  |   |   |  |  |
|                                                                                       |   |   |   |   |   |   |  |   |   |  |  |
| <b>7.0 Scenario Analysis</b>                                                          |   |   |   |   |   |   |  |   |   |  |  |
| 7.1 Define and run baseline scenario reflecting current campus energy conditions      | A | R | C | C | I | C |  | C | C |  |  |
| 7.2 Define and run alternative investment scenarios across technology combinations    | A | R | R | C | I |   |  | C | C |  |  |
| 7.3 Execute sensitivity analysis on critical input assumptions                        | A | R | R | C | I |   |  | C |   |  |  |
| 7.4 Compile internal working documentation of scenario results and trade-off findings | R | C | C | C | I |   |  |   |   |  |  |
|                                                                                       |   |   |   |   |   |   |  |   |   |  |  |
| <b>8.0 Validation and Quality Assurance</b>                                           |   |   |   |   |   |   |  |   |   |  |  |

|                                                                                   |   |   |   |   |   |   |   |   |   |   |   |
|-----------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|
| 8.1 Validate all model outputs against reference datasets and industry benchmarks | I | R | C | C | I | C |   | R | C | A |   |
| 8.2 Test integrated framework for edge cases and boundary conditions              | I | R |   |   | I |   | C |   |   | A |   |
| 8.3 Review framework usability for non-technical stakeholders                     | R | C |   | C | I | R |   |   |   | C | I |
| 8.4 Audit input documentation and assumption traceability                         | R | C | C | C | I |   |   |   |   | A |   |
| 8.5 Address validation findings and conduct final internal review                 | R | R | C | C | I |   |   | C |   | A |   |
|                                                                                   |   |   |   |   |   |   |   |   |   |   |   |
| <b>9.0 Reporting and Delivery</b>                                                 |   |   |   |   |   |   |   |   |   |   |   |
| 9.1 Write user guide and finalize methodology                                     | R | C | C | C | I |   |   |   |   |   |   |



|                                                                            |   |   |   |   |   |   |  |   |   |  |   |
|----------------------------------------------------------------------------|---|---|---|---|---|---|--|---|---|--|---|
| documentation                                                              |   |   |   |   |   |   |  |   |   |  |   |
| 9.2 Prepare scenario comparison summary tables and visualizations          | A | R | C | C | I |   |  |   |   |  |   |
| 9.3 Synthesize findings into structured client-facing decision report      | R | C | R | C | A | C |  |   |   |  |   |
| 9.4 Draft investment strategy recommendations aligned to campus objectives | R | C | R | R | A | C |  | C | C |  |   |
| 9.5 Incorporate final stakeholder feedback into decision report            | R | C | C | C | A | C |  |   |   |  | I |
|                                                                            |   |   |   |   |   |   |  |   |   |  |   |
| <b>10.0 Project Closure</b>                                                |   |   |   |   |   |   |  |   |   |  |   |
| 10.1 Prepare and rehearse final project presentation                       | R | C | C | C | A |   |  |   |   |  | I |
| 10.2 Submit final report and obtain formal project                         | R |   |   |   | A |   |  |   |   |  | R |

|                         |  |  |  |  |  |  |  |  |  |  |  |
|-------------------------|--|--|--|--|--|--|--|--|--|--|--|
| acceptance from sponsor |  |  |  |  |  |  |  |  |  |  |  |
|-------------------------|--|--|--|--|--|--|--|--|--|--|--|

R = Responsible | A = Accountable | C = Consulted | I = Informed

## APPENDIX D: Budget Justification

Table D-1: Budget Justification by Phase and Task

| ID           | TASKS                                                  | PEOPLE | HOURS | RATE/HR | EST LABOUR COST | MATERIALS | MISC | TOTAL TASK COST |
|--------------|--------------------------------------------------------|--------|-------|---------|-----------------|-----------|------|-----------------|
| <b>1.0</b>   | <b>Project Initiation</b>                              |        |       |         |                 |           |      |                 |
| <b>1.1</b>   | Conduct project kick-off meeting                       |        |       |         |                 |           |      |                 |
| <b>1.1.1</b> | Define project scope, boundaries, and success criteria | 7      | 13    | 162.5   | 2112.5          |           |      |                 |

|       |                                                                       |    |    |       |        |  |  |  |
|-------|-----------------------------------------------------------------------|----|----|-------|--------|--|--|--|
| 1.1.2 | Identify and map stakeholders                                         | 10 | 8  | 208.5 | 1668   |  |  |  |
| 1.2   | Develop project charter                                               | 4  | 21 | 137   | 2877   |  |  |  |
| 1.3   | Establish communication plan and reporting cadence                    | 9  | 8  | 162.5 | 1300   |  |  |  |
| 1.4   | Obtain project sponsor approval and sign-off                          | 2  | 4  | 117   | 468    |  |  |  |
|       |                                                                       |    |    |       |        |  |  |  |
| 2.0   | <b>Project Planning</b>                                               |    |    |       |        |  |  |  |
| 2.1   | Requirements Definition                                               |    |    |       |        |  |  |  |
| 2.1.1 | Conduct stakeholder interviews to gather decision-making requirements | 7  | 16 | 141.5 | 2264   |  |  |  |
| 2.1.2 | Define evaluation criteria and framework objectives                   | 8  | 13 | 158.5 | 2060.5 |  |  |  |
| 2.1.3 | Document initial assumptions, constraints, and exclusions             | 6  | 9  | 127   | 1143   |  |  |  |
| 2.1.4 | Obtain stakeholder sign-off on requirements                           | 4  | 4  | 133.5 | 534    |  |  |  |
| 2.2   | Research and Background Analysis                                      |    |    |       |        |  |  |  |
| 2.2.1 | Review literature on renewable energy technologies                    | 6  | 25 | 138.5 | 3462.5 |  |  |  |

|       |                                                                   |   |    |       |        |  |  |  |
|-------|-------------------------------------------------------------------|---|----|-------|--------|--|--|--|
| 2.2.2 | Identify and catalog data sources and regulatory incentives       | 9 | 16 | 185.5 | 2968   |  |  |  |
| 2.3   | Project Management Planning                                       |   |    |       |        |  |  |  |
| 2.3.1 | Develop detailed project schedule and milestones                  | 6 | 16 | 157   | 2512   |  |  |  |
| 2.3.2 | Assign roles and responsibilities (RACI)                          | 5 | 13 | 146   | 1898   |  |  |  |
| 2.3.3 | Develop risk management plan                                      | 7 | 16 | 171   | 2736   |  |  |  |
| 2.3.4 | Develop quality management plan                                   | 5 | 13 | 177   | 2301   |  |  |  |
|       |                                                                   |   |    |       |        |  |  |  |
| 3.0   | <b>Framework Design</b>                                           |   |    |       |        |  |  |  |
| 3.1   | Design framework architecture and data flow between components    | 5 | 21 | 127.5 | 2677.5 |  |  |  |
| 3.2   | Establish input/output specifications for each model              | 8 | 16 | 156.5 | 2504   |  |  |  |
| 3.3   | Design scenario comparison and sensitivity analysis structure     | 6 | 16 | 130.5 | 2088   |  |  |  |
| 3.4   | Define Excel workbook structure, layout, and data entry standards | 4 | 13 | 113.5 | 1475.5 |  |  |  |

|       |                                                                                      |   |    |       |        |  |  |  |
|-------|--------------------------------------------------------------------------------------|---|----|-------|--------|--|--|--|
| 3.5   | Obtain stakeholder review and approval of framework design                           | 7 | 8  | 162.5 | 1300   |  |  |  |
|       |                                                                                      |   |    |       |        |  |  |  |
| 4.0   | <b>Data Acquisition and Preparation</b>                                              |   |    |       |        |  |  |  |
| 4.1   | Collect and validate historical campus energy consumption data                       | 5 | 25 | 122.5 | 3062.5 |  |  |  |
| 4.2   | Collect and validate site-specific renewable resource data (solar, wind, geothermal) | 8 | 25 | 173   | 4325   |  |  |  |
| 4.3   | Collect capital cost estimates and O&M benchmarks per technology                     | 7 | 16 | 211.5 | 3384   |  |  |  |
| 4.4   | Obtain utility tariff structures, rate projections, and applicable incentives        | 6 | 13 | 148.5 | 1930.5 |  |  |  |
| 4.5   | Clean and preprocess all collected data for model input                              | 4 | 25 | 123.5 | 3087.5 |  |  |  |
|       |                                                                                      |   |    |       |        |  |  |  |
| 5.0   | <b>Model Development</b>                                                             |   |    |       |        |  |  |  |
| 5.1   | Campus Energy Demand Forecasting Model                                               |   |    |       |        |  |  |  |
| 5.1.1 | Build baseline demand model                                                          | 7 | 41 | 157.5 | 6457.5 |  |  |  |

|       |                                                                                   |   |    |       |        |  |  |  |
|-------|-----------------------------------------------------------------------------------|---|----|-------|--------|--|--|--|
|       | incorporating historical data, seasonal variation, and occupancy patterns         |   |    |       |        |  |  |  |
| 5.1.2 | Develop near-term and long-term demand projections                                | 7 | 25 | 106.5 | 2662.5 |  |  |  |
| 5.2   | Renewable Energy Production Estimation Model                                      |   |    |       |        |  |  |  |
| 5.2.1 | Build generation estimation logic for solar PV, wind, and geothermal systems      | 7 | 41 | 162   | 6642   |  |  |  |
| 5.2.2 | Build battery storage dispatch and capacity model                                 | 7 | 32 | 162   | 5184   |  |  |  |
| 5.3   | Cost-Benefit Analysis Model                                                       |   |    |       |        |  |  |  |
| 5.3.1 | Build capital expenditure, O&M cost, and utility savings structure                | 6 | 25 | 147.5 | 3687.5 |  |  |  |
| 5.3.2 | Develop NPV, IRR, and payback period calculation engine                           | 6 | 32 | 96.5  | 3088   |  |  |  |
| 5.4   | Environmental Impact Assessment Model                                             |   |    |       |        |  |  |  |
| 5.4.1 | Establish campus emissions baseline and reduction estimation logic per technology | 6 | 25 | 90.5  | 2262.5 |  |  |  |
| 5.4.2 | Align outputs with Northeastern sustainability                                    | 4 | 16 | 128.5 | 2056   |  |  |  |

|            |                                                                                |   |    |       |        |  |  |  |
|------------|--------------------------------------------------------------------------------|---|----|-------|--------|--|--|--|
|            | reporting standards                                                            |   |    |       |        |  |  |  |
|            |                                                                                |   |    |       |        |  |  |  |
| <b>6.0</b> | <b>Framework Integration</b>                                                   |   |    |       |        |  |  |  |
| <b>6.1</b> | Link all model outputs into unified decision-support environment               | 7 | 25 | 155.5 | 3887.5 |  |  |  |
| <b>6.2</b> | Build scenario configuration interface for portfolio definition                | 4 | 21 | 123.5 | 2593.5 |  |  |  |
| <b>6.3</b> | Build and test sensitivity analysis module                                     | 6 | 25 | 146.5 | 3662.5 |  |  |  |
| <b>6.4</b> | Build scenario comparison dashboard                                            | 7 | 25 | 142.5 | 3562.5 |  |  |  |
| <b>6.5</b> | Conduct mid-project stakeholder review and incorporate feedback                | 7 | 9  | 162.5 | 1462.5 |  |  |  |
|            |                                                                                |   |    |       |        |  |  |  |
| <b>7.0</b> | <b>Scenario Analysis</b>                                                       |   |    |       |        |  |  |  |
| <b>7.1</b> | Define and run baseline scenario reflecting current campus energy conditions   | 8 | 16 | 158.5 | 2536   |  |  |  |
| <b>7.2</b> | Define and run alternative investment scenarios across technology combinations | 7 | 25 | 179   | 4475   |  |  |  |
| <b>7.3</b> | Execute sensitivity analysis on critical input assumptions                     | 6 | 21 | 162   | 3402   |  |  |  |

|     |                                                                                   |   |    |       |        |  |  |  |
|-----|-----------------------------------------------------------------------------------|---|----|-------|--------|--|--|--|
| 7.4 | Compile internal working documentation of scenario results and trade-off findings | 5 | 16 | 83    | 1328   |  |  |  |
|     |                                                                                   |   |    |       |        |  |  |  |
| 8.0 | <b>Validation and Quality Assurance</b>                                           |   |    |       |        |  |  |  |
| 8.1 | Validate all model outputs against reference datasets and industry benchmarks     | 9 | 25 | 212.5 | 5312.5 |  |  |  |
| 8.2 | Test integrated framework for edge cases and boundary conditions                  | 5 | 21 | 118.5 | 2488.5 |  |  |  |
| 8.3 | Review framework usability for non-technical stakeholders                         | 7 | 9  | 139   | 1251   |  |  |  |
| 8.4 | Audit input documentation and assumption traceability                             | 6 | 13 | 123   | 1599   |  |  |  |
| 8.5 | Address validation findings and conduct final internal review                     | 7 | 16 | 175.5 | 2808   |  |  |  |
|     |                                                                                   |   |    |       |        |  |  |  |
| 9.0 | <b>Reporting and Delivery</b>                                                     |   |    |       |        |  |  |  |
| 9.1 | Write user guide and finalize methodology documentation                           | 5 | 25 | 83    | 2075   |  |  |  |
| 9.2 | Prepare scenario comparison summary tables and                                    | 5 | 21 | 116.5 | 2446.5 |  |  |  |



|             |                                                                        |   |    |       |         |  |  |  |
|-------------|------------------------------------------------------------------------|---|----|-------|---------|--|--|--|
|             | visualizations                                                         |   |    |       |         |  |  |  |
| <b>9.3</b>  | Synthesize findings into structured client-facing decision report      | 6 | 25 | 188.5 | 4712.5  |  |  |  |
| <b>9.4</b>  | Draft investment strategy recommendations aligned to campus objectives | 8 | 16 | 251   | 4016    |  |  |  |
| <b>9.5</b>  | Incorporate final stakeholder feedback into decision report            | 7 | 9  | 162.5 | 1462.5  |  |  |  |
|             |                                                                        |   |    |       |         |  |  |  |
| <b>10.0</b> | <b>Project Closure</b>                                                 |   |    |       |         |  |  |  |
| <b>10.1</b> | Prepare and rehearse final project presentation                        | 6 | 16 | 151.5 | 2424    |  |  |  |
| <b>10.2</b> | Submit final report and obtain formal project acceptance from sponsor  | 3 | 4  | 166.5 | 666     |  |  |  |
|             | TOTAL EST LABOUR COST                                                  |   |    |       | 146,350 |  |  |  |

## APPENDIX E: Resource Allocation

Table E-1: Resource Allocation Plan (All Project Roles)

| Role Group                     | Role Title                       | Assigned To          | Rate/Hr | Effort in Hours   |
|--------------------------------|----------------------------------|----------------------|---------|-------------------|
|                                | Project Manager                  | Nadukuru, J.S.R.     | \$50    | 474.2             |
|                                | Financial Analyst                | Lokesh, A.           | \$45    | 238.8             |
|                                | Data & Systems Analyst           | Sudhandrakumar, S.K. | \$55    | 564.5             |
|                                | IT/Data Engineer                 | IT                   | \$75    | 170.2             |
|                                | Sustainability Analyst           | Senthilvel, S.       | \$45    | 205.4             |
|                                | Quality Assurance Analyst (QA)   | To Be Assigned       | \$50    | 72.2              |
|                                | Renewable Energy Engineer        | To Be Assigned       | \$70    | 228.7             |
|                                | Northeastern Admin & Energy SPOC | To Be Assigned       | \$55    | 12.8              |
|                                | Facilities & Energy Manager      | To Be Assigned       | \$55    | 131               |
|                                | Energy Consultant                | To Be Assigned       | \$85    | 100.9             |
|                                | Project Sponsor                  | To Be Assigned       | \$90    | 252.6             |
| <b>Grand Total (All Roles)</b> |                                  |                      |         | <b>2451.3 hrs</b> |
|                                |                                  |                      |         |                   |
|                                |                                  |                      |         |                   |

Note: Roles marked 'To Be Assigned' are defined in the project stakeholder mapping and will be formally recruited before project execution begins. Rates reflect standard market rates for each role. Government / Regulatory Bodies and Campus Community are engagement-only roles with no direct labor cost. The Grand Total includes all billable roles across the full project lifecycle.

